

Enhanced Learning Center Display System - Documentation

Overview

The Enhanced Learning Center Display System is a comprehensive web application designed for educational institutions, conference centers, and presentation environments. This version includes advanced time and weather configuration capabilities with real location-based data for North American cities.

New Features (Version 2.0)



Real Location-Based Weather System

- **75+ North American Cities:** Comprehensive database including US, Canadian, and Mexican cities
- **Real Coordinates:** Each city uses actual latitude/longitude for accurate weather data
- **Live Weather Data:** Real-time temperature, conditions, humidity, and wind information
- **Smart City Selection:** Searchable dropdown with properly formatted city names



Advanced Time Configuration

- **25+ Timezones:** Complete North American timezone support with UTC offsets
- **Multiple Formats:** 12-hour/24-hour time display options
- **Date Formatting:** Five different date format options (Full, Long, Medium, Short, ISO)
- **Visual Customization:** Eight font size options from 14px to 48px
- **Display Controls:** Toggle seconds and timezone information visibility

Technical Architecture

Frontend Framework

- **React 18** with Vite build system
- **Tailwind CSS** for responsive styling
- **Shadcn/UI** components for professional interface
- **Lucide React** icons for consistent iconography

Data Sources

- **Open-Meteo API:** Free weather data service (no API key required)
- **Browser Timezone API:** Automatic timezone detection and formatting
- **LocalStorage:** Settings persistence across sessions

Key Components

Core Application Structure

Plain Text

```
src/
├── components/
│   ├── Navigation.jsx           # Main navigation component
│   ├── PresentationPage.jsx     # Main display interface
│   ├── UploadPage.jsx          # File management interface
│   ├── SettingsPage.jsx        # Configuration interface
│   ├── FileDisplay.jsx         # File rendering component
│   └── widgets/
│       ├── TimeWidget.jsx       # Enhanced time display
│       └── WeatherWidget.jsx    # Enhanced weather display
├── contexts/
│   ├── FileContext.jsx         # File management state
│   └── SettingsContext.jsx      # Enhanced settings state
├── data/
│   ├── cities.js               # North American cities database
│   └── timezones.js            # Timezone and formatting data
└── App.jsx                     # Main application component
```

Installation & Setup

Prerequisites

- Node.js 18+ and pnpm package manager
- Modern web browser with JavaScript enabled
- Internet connection for weather data

Local Development Setup

1. Clone or Extract Files
2. Install Dependencies

3. Start Development Server

4. Build for Production

Production Deployment

Static Hosting (Recommended)

1. Build the application: `pnpm run build`
2. Upload `dist/` contents to your web server
3. Configure server for SPA routing (redirect all routes to `index.html`)

Docker Deployment

Plain Text

```
FROM nginx:alpine
COPY dist/ /usr/share/nginx/html/
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 80
```

Configuration Guide

Time Configuration Options

Timezone Selection

- **US Timezones:** Eastern, Central, Mountain, Pacific, Alaska, Hawaii
- **Canadian Timezones:** Atlantic, Newfoundland, plus regional variations
- **Mexican Timezones:** Central, Pacific, Eastern (Mexico), Mountain
- **Special Cases:** Arizona (no DST), Saskatchewan (no DST)

Time Display Formats

- **12-hour:** 3:45:30 PM (with AM/PM indicator)
- **24-hour:** 15:45:30 (military time format)

Date Display Formats

- **Full:** Monday, September 22, 2025
- **Long:** September 22, 2025

- **Medium:** Sep 22, 2025
- **Short:** 9/22/2025
- **ISO:** 2025-09-22

Font Size Options

- Small (14px), Medium (16px), Large (18px), Extra Large (20px)
- 2X Large (24px), 3X Large (30px), 4X Large (36px), 5X Large (48px)

Weather Configuration Options

Supported Cities by Region

United States (50 cities)

- Major metros: New York, Los Angeles, Chicago, Houston, Phoenix
- Regional centers: Atlanta, Denver, Seattle, Miami, Boston
- State capitals and significant cities across all states

Canada (25 cities)

- Provincial capitals: Toronto, Montreal, Vancouver, Calgary
- Major centers: Ottawa, Winnipeg, Quebec City, Halifax
- Regional cities across all provinces

Mexico (15 cities)

- Major cities: Mexico City, Guadalajara, Monterrey
- Border cities: Tijuana, Juárez, Mexicali
- Tourist destinations: Cancún, Mérida

Weather Data Features

- **Real-time Updates:** Data refreshes every 10 minutes
- **Comprehensive Info:** Temperature, conditions, humidity, wind
- **Weather Icons:** Dynamic icons based on current conditions
- **Temperature Units:** Celsius or Fahrenheit selection

Usage Instructions

Basic Operation

1. Upload Content

- Navigate to Upload page
- Drag and drop files or use file selector
- Supported formats: PDF, images (JPG, PNG), videos (MP4, WebM)
- Add web links for online content

2. Configure Settings

- Access Settings page for all configuration options
- Set presentation timing and loop behavior
- Configure time display preferences
- Select weather location and units

3. Start Presentation

- Return to Presentation page
- Content displays automatically based on settings
- Real-time widgets update continuously

Advanced Features

Presentation Control

- **Auto Play:** Automatic progression through content
- **Loop Modes:** Continuous, count-based, or time-based stopping
- **Timing Control:** Configurable display duration per file

Widget Customization

- **Time Widget:** Timezone, format, font size, visibility options
- **Weather Widget:** Location selection, temperature units
- **Layout:** Fixed sidebar design matching original mockup

API Integration

Weather Data (Open-Meteo)

JavaScript

```
// Example API call structure
const weatherUrl = `https://api.open-meteo.com/v1/forecast?
latitude=${lat}&longitude=${lon}&current=temperature_2m,weather_code,relative
_humidity_2m,wind_speed_10m&temperature_unit=${unit}&timezone=auto`
```

Timezone Handling

JavaScript

```
// Example timezone formatting
const timeString = date.toLocaleTimeString('en-US', {
  timeZone: 'America/New_York',
  hour12: false,
  hour: 'numeric',
  minute: '2-digit',
  second: '2-digit'
})
```

Troubleshooting

Common Issues

Weather Data Not Loading

- Check internet connection
- Verify city selection in settings
- Weather API may have temporary outages

Time Display Issues

- Ensure browser supports Intl.DateTimeFormat
- Check timezone selection in settings
- Verify system clock accuracy

File Upload Problems

- Check file size limits (browser dependent)
- Verify supported file formats
- Clear browser cache if needed

Performance Optimization

Large File Handling

- Optimize images before upload
- Use appropriate video compression
- Consider file size limits for smooth playback

Memory Management

- Clear unused files regularly
- Restart application for long-running displays
- Monitor browser memory usage

Browser Compatibility

Supported Browsers

- **Chrome/Chromium:** 90+ (recommended)
- **Firefox:** 88+
- **Safari:** 14+
- **Edge:** 90+

Required Features

- ES2020 JavaScript support
- CSS Grid and Flexbox
- Fetch API and Promises
- Local Storage API

Security Considerations

Data Privacy

- All settings stored locally in browser
- No user data transmitted to external servers
- Weather data requests are anonymous

Content Security

- File uploads processed client-side only

- No server-side file storage
- XSS protection through React's built-in sanitization

Maintenance & Updates

Regular Maintenance

- Update dependencies quarterly
- Monitor weather API status
- Test timezone changes during DST transitions
- Verify city database accuracy

Version Control

- Use semantic versioning (current: 2.0.0)
- Tag releases for stable deployments
- Maintain changelog for updates

Support & Resources

Documentation

- Component API documentation in source files
- Inline comments for complex logic
- README files for setup instructions

Development Resources

- React documentation: <https://react.dev>
- Tailwind CSS: <https://tailwindcss.com>
- Open-Meteo API: <https://open-meteo.com>

License & Credits

Third-Party Services

- **Open-Meteo:** Weather data (CC BY 4.0)
- **Lucide:** Icon library (ISC License)

- **Tailwind CSS:** Styling framework (MIT License)

Attribution

Enhanced Learning Center Display System developed for educational and presentation environments with comprehensive North American location support.

Last Updated: September 22, 2025

Version: 2.0.0